

Repeated Games

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Stage Game and Repeated Game

- Let $\Gamma = \{I, S, u\}$ be a simultaneous move game.
- A repeated game of $T + 1$ periods denoted by $\Gamma^T(\delta)$ is a repetition of the stage game $T + 1$ times $(0, \dots, T)$:
 - players observe history $h^t = \{a^0, a^1, \dots, a^{t-1}\}$ in period t .
 - the present value at date 0 of the payoffs received in period t is discounted by $\delta^t \in [0, 1]$.
- A pure strategy for player i is $\{s_i^t\}_{t=0}^T$ that maps $h^t \in H^t$ to action $a_i \in A_i$.
- A mixed (behavior) strategy σ_i is a sequence of maps $\{\sigma_i^t\}_{t=0}^T$ that maps the set of histories H^t to mixed actions $\alpha_i^t \in \times_{h^t \in H^t} \Delta A(h^t)$.
- Each α_i^t depends only on h^t , not on past mixing $\{\alpha_{-i}^\tau\}_{\tau=0}^t$ of the other players.
- A new subgame begins in each period.

Finitely Repeated Games

- The game ends with probability 1 in round $T < \infty$.
- We can solve for SPNE using backward induction starting from round T .

Finitely Repeated Prisoner's Dilemma

		Player 2	
		C	NC
Player 1	C	$-9, -9$	$0, -10$
	NC	$-10, 0$	$-1, -1$

Figure: Prisoner's dilemma 1

Finitely Repeated Prisoner's Dilemma

- In round $T + 1$, the NE of the subgame is (C, C) .
- Now round T , (NC, NC) cannot be an equilibrium of the subgame starting at this period since:

$$EU_1(NC, NC) = -1 - 9\delta < 0 - 9\delta = EU_1(C, NC)$$

- Similarly, (NC, C) and (C, NC) cannot be an equilibrium as well.
- Folding the game back, we can show that confess is the unique NE of every subgame.
- Therefore, the unique SPNE is for both to confess in each round.
- The main problem is that a deviation in the last round cannot be punished.

General Observation Finitely Repeated Games

Proposition 1

Suppose that the simultaneous move game Γ has a unique NE σ^* . If $T < \infty$, then the unique SPNE of $\Gamma^T(\delta)$ is $\alpha_i^{t,*} = \sigma_i^*$.

Proof

The proof follows an inductive argument. Since $\Gamma^0(\delta) = \Gamma$, this result holds trivially for $T = 0$. Suppose that it holds for a game of length $T = k - 1$ (k rounds). We have to show that it holds for $T = k$ ($k + 1$ rounds). By hypothesis, the subgame starting in round 1 has a unique SPNE of $\alpha_i^{t,*} = \sigma_i^*$ for all $t = 1, \dots, k$. Therefore, at round 0, player i solves

$$\max_{\sigma_i} u_i(\sigma_i, \sigma_{-i}) + \sum_{t=1}^k \delta^t u_i(\sigma_i^*, \sigma_{-i}^*)$$

General Observation for Finitely Repeated Games

Proof continued

Note $\sum_{t=1}^k \delta^t = \delta \sum_{t=0}^{k-1} \delta^t = \delta \frac{1-\delta^k}{1-\delta}$.

The optimization problem for i can be rewritten as:

$$\max_{\sigma_i} u_i(\sigma_i, \sigma_{-i}) + \delta \frac{1-\delta^k}{1-\delta} u_i(\sigma_i^*, \sigma_{-i}^*)$$

The last term is just a constant. Thus, the solution is $r_i(\sigma_{-i})$ leading to a unique NE of σ^* .

Multiple NE of the stage game

- Then, SPNE may involve a play that is not NE of the stage game.

		Player 2		
		C	NC	P
Player 1	C	$-9, -9$	$0, -10$	$-15, -10$
	NC	$-10, 0$	$-1, -1$	$-15, -10$
	P	$-10, -15$	$-10, -15$	$-12, -12$

Figure: Prisoner's dilemma 2

- Suppose that this game is repeated twice. Can (NC, NC) be supported in the first round? What should the discount factor be?

Supergames

- A Super game $\Gamma^\infty(\delta)$ is an infinitely repeated stage game.
 - δ may be interpreted as the probability that the game will end in each period.
- Backward induction is not possible since there is no last stage.

Upper-hemicontinuity at $T = \infty$

Suppose that the simultaneous move game Γ has a unique NE σ^* . Then $\alpha_i^{t,*} = \sigma_i^*$ for all i and t is also a SPNE of $\Gamma^\infty(\delta)$ for $\delta < 1$.

- Are there other SPNE that perform better?

Minmax values

- Supporting a strategy profile that is not a NE of the stage game requires the ability to punish deviators.
- Minmax strategy is a way to push deviators.

Minmax strategy profile (m_i^i, m_{-i}^i)

$$(m_i^i, m_{-i}^i) = \operatorname{argmin}_{\alpha_{-i}} \left[\max_{\alpha_i} u_i(\alpha_i, \alpha_{-i}) \right]$$

Minmax payoff

$$\underline{v}_i = \min_{\alpha_{-i}} \left[\max_{\alpha_i} u_i(\alpha_i, \alpha_{-i}) \right]$$

Example

		Pauper	
		W	S
Gov.	A	3, 2	-1, 3
	NA	-1, 1	0, 0

Figure

- Let θ_a denote the probability of A in a mixed strategy by the government.
- Let γ_w denote the probability of W in a mixed strategy by the worker.
- $(\theta_a, \gamma_w) = (\frac{1}{2}, \frac{1}{5})$ is the unique mixed strategy equilibrium of the stage game.

Minmax values

- What are the minmax strategies (m_g^p, m_p^g) and values $(\underline{v}_g, \underline{v}_p)$?

(Assist, Work) in the long run

- Consider the following 'trigger' strategy in period t by the government in the infinitely repeated game:

$$a_g^0 = A \qquad a_g^t = \begin{cases} A & \text{if } a_p^k = W \ \forall k = \{0, 1, \dots, t-1\} \\ m_g^p & \text{otherwise} \end{cases}$$

- What is the minimum $\underline{\delta}$ required to induce the pauper to work in every subgame?

Folk Theorem

Proposition 2

For every feasible payoff vector v with $v_i > \underline{v}_i$ for all players, there exists $\underline{\delta}$ such that $\forall \delta \in (\underline{\delta}, 1)$, there is a Nash equilibrium of $\Gamma^\infty(\delta)$ with payoff v .

- Let v correspond to some strategy profile a .
- The proof involves specifying a trigger strategy for each player i :
 - play a_i corresponding to v if all players played according to a or two or more players have deviated in earlier periods;
 - play according to minmax strategy for player j if player j has deviated unilaterally in an earlier period.
- Why Nash equilibrium and not SPNE?
 - The minmax strategy profile need not be a NE of the stage game.
 - In the Gov-Pauper game, (NA, W) is not a Nash equilibrium of the subgame following a deviation by the pauper.

(Perfect) Folk Theorem

Proposition 3

Suppose that σ^* is a Nash equilibrium of the simultaneous-move game Γ and consider the supergame $\Gamma^\infty(\delta)$ with $\delta < 1$. Let σ be a one shot strategy that is strictly preferred by all players (i.e., $u_i(\sigma) > u_i(\sigma^*) \forall i$). There exists $\underline{\delta} < 1$ such that if $\delta \in [\underline{\delta}, 1)$, then the following strategies are a SPNE of $\Gamma^\infty(\delta)$. For all i ,

$$\sigma_i^0 = \sigma_i \qquad \sigma_i^t = \begin{cases} \sigma_i & \text{if } h_t = \{\sigma, \sigma, \dots, \sigma\} \\ \sigma_i^* & \text{otherwise} \end{cases}$$

- If σ is a mixed strategy, we require that the mixing is ex-post verifiable.

References

- Fudenberg and Tirole- Chapter 5